

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed, developed, and optimized large-scale **AI model pre-training pipelines** on distributed GPU infrastructures, managing data flow for experiments.
- Researched and prototyped advanced **transformer architectures** for large language models, focusing on efficiency and scalability for new AI systems.
- Conducted extensive **experimental analysis** on training methodologies for large-scale models, iterating to improve computational efficiency by approximately **18%**.
- Identified and resolved complex bottlenecks in **distributed training systems**, enhancing performance and reliability across large compute environments.
- Contributed to data curation strategies and baseline improvements, strengthening model training outcomes for **multi-modal AI systems**.
- Collaborated with international engineering and research teams to accelerate innovation and deliver high-impact **AI capabilities**.

SKILLS

AI & ML Frameworks – PyTorch, TensorFlow, JAX, scikit-learn

Model Architectures – Large Language Models, Transformer Architectures, Multi-modal AI

Distributed Systems – Large-Scale GPU Training, Parallel Training, Distributed Machine Learning

Programming & Scripting – Python, Java, SQL, Bash

Data Engineering & Analytics – Data Curation, ETL, Data Pipelines, Pandas, NumPy, R

Cloud & Infrastructure – AWS (EC2, S3, Lambda), Docker, Kubernetes

MLOps & Tools – MLOps, Jenkins, Git, Jira, Confluence, Splunk, Dynatrace

PROJECTS

Large-Scale Model Pre-training Pipeline

- Engineered a Python-based data pipeline on **AWS S3 and Lambda** to prepare and feed large datasets for transformer model pre-training.
- Implemented distributed training strategies using **PyTorch DistributedDataParallel**, enabling model training across multiple GPU nodes and improving throughput by an estimated **22%**.
- Developed monitoring and logging mechanisms using **Splunk** to track experiment progress and identify performance bottlenecks in near real-time.

AI Model Experimentation Framework

- Designed and implemented a Python framework for conducting systematic research experiments on novel **AI architectures**, managing hyperparameter tuning and result aggregation.
- Trained a deep learning model (**scikit-learn**) on a synthetic dataset of 150,000 records to classify data patterns, achieving an accuracy of **0.89**.
- Integrated experiment tracking with **Jira and Confluence** for seamless collaboration with research team members.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)